Qist Management Software: Documentation

Project: Electronics shop ke liye qist (installment) management system **Version:** 2.2 (Draft). 2.0 par base, 5.6 mein exact formulas, 5.7 mein costing aur profit/loss, aur Final Specification (docx) ke review ke baad expenses, idempotency, acceptance checklist wagaira add

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1. Introduction

Ye software electronics dukaan ke qist ka poora hisaab rakhta hai: customer ka record, product ki qist par farokht, har qist ki tareekh, wasooli, baqaya raqam aur reports. Maqsad ye hai ke dukandar kam se kam typing aur kam se kam waqt mein sab kuch sanbhal sake.

Maqasid

- Customer aur guarantor ka mukammal record
- Qist plan khud ba khud generate hona
- Ek click mein qist wasool karna aur receipt dena
- Aaj ki due aur late qiston ka foran pata chalna
- Rozana aur mahana reports

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2. Technology Stack

Hissa	Technology
Backend	ASP.NET Core 10 Web API
Data Access	**Dapper + Stored Procedures** (EF Core istemal nahi hoga)
Business Logic	**SQL Stored Procedures mein**, C# sirf call karta hai
Database	Microsoft SQL Server **Developer Edition** (development/testing ke liye)
Frontend	HTML5, Bootstrap 5, CSS, **jQuery + AJAX**
Authentication	JWT, Roles + Permissions
Hosting	Frontend `wwwroot` mein, ek hi project se chalega

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3. Users aur Roles

Role	Ikhtiyarat
Owner (Admin)	Sab kuch: users, reports, plans delete/edit, settings
Staff	Customer add, naya plan, qist wasool, receipt print
Accountant (optional)	Reports, purchase, stock

Ye default roles hain. Asal ikhtiyarat **Permissions** se tay hote hain, Owner jis role ko jo ijazat chahe de sakta hai (section 6.2).

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4. Modules aur Features

4.1 Dashboard

- Aaj ki kul wasooli
- Aaj due qistein (count aur raqam)
- Late qistein (laal rang mein)
- Kul baqaya raqam
- Is mahine ki farokht

4.2 Customers

- Add, edit, search (naam, phone, CNIC se)
- Customer ki detail page par: tamam plans, baqaya, payment history
- Guarantor ki details (ek ya zyada)
- Customer ka status: Active / Blacklisted

4.3 Products

- Product ka naam, **brand**, category, nagad qeemat
- Brand/category ke hisaab se markup khud lagti hai (5.4 dekhein)
- Stock `StockTransactions` se nikalta hai (4.10), product mein seedha nahi likha jata

4.4 Naya Qist Plan (Sale)

- Sale ki qism: **Cash** ya **Installment** (qist par hi contract aur qistein banti hain)
- Customer select karna (cash sale mein zaroori nahi)
- **Ek plan mein ek ya zyada products** (jaise TV + fridge) cart ki tarah add karna, har product ki qeemat, tadaad (qty) aur serial/IMEI ke saath. Sab products ki qeemat jama hokar **ek hi plan aur ek hi qist schedule** banta hai
- Advance raqam, qiston ki tadaad, shuru ki tareekh
- **Qist ka waqfa (frequency) customize:** Hafta war, 15 din, Mahana, ya Custom (har N din)
- **Markup (munafa) customize:** koi nahi / fixed raqam / percentage / percentage per month / ya khud final qeemat likh dein (5.2 dekhein)
- **Auto + Manual dono:** system khud hisaab lagata hai, aur staff kisi bhi raqam ko apne hisaab se badal sakta hai (baqi raqamein khud update hoti hain)
- **Preview jo edit ho sakta hai:** save se pehle qiston ki list dikhti hai, har qist ki date aur raqam badli ja sakti hai
- Qist ki raqam ko gol karna (jaise 10/50/100 tak), farq aakhri qist mein
- Plan save hotay hi qistein ban jati hain

4.5 Qist Wasooli (Payments)

- Customer search karke uske pending plans dekhna

- Raqam likhna aur "Jama Karo" dabana
- Partial payment (adhi qist) ka support
- Receipt number khud ban jaye, print aur WhatsApp ka option

4.6 Reports

- Rozana wasooli
- Mahana wasooli aur farokht
- Defaulters (late customers) ki list
- Customer wise baqaya
- **Rate List:** product wise, har plan (3/6/12 mahine) ka kul qeemat aur mahana qist, print / PDF / WhatsApp
- Print aur Excel export

4.7 Settings

- **Shop Profile** (4.13): dukaan ka naam, logo, address, phone, receipt ka header/footer, sab Owner kabhi bhi badal sakta hai
- Markup Rules ka intezam (brand/category wise, plan ki muddat wise),
- Default frequency, default markup type/value, markup kis par lage (poori qeemat ya baqaya), qist round-off (0/10/50/100 aur optional +1), late fee (optional)
- Language: Roman Urdu / English (Urdu RTL baad mein)

4.8 Branches aur Warehouses

- Dukaan ki ek ya zyada shakhain, har ki alag warehouse/showroom
- Receipt/sale numbers branch ke hisaab se (jaise `LHR-R-2026-000123`), user ek branch se jura
- **Ek hi database, ek hi shop:** sab branches ka data ek jagah, har record par `BranchId`
- **Customer sab branches mein ek hi** (ek customer kisi bhi branch se khareed ya qist de sakta hai)
- **Plan us branch ka** jahan bana, magar qist **kisi bhi branch mein jama** ho sakti hai. Payment par us branch ka naam likha jata hai jahan raqam li gayi
- **Stock har warehouse ka alag**, aur ek branch se doosri ko stock transfer (`StockTransactions` mein Transfer)
- **Reports:** ek branch ke ya sab branches ke mila kar. Owner sab dekhta hai, branch ka staff sirf apni branch ka (permission se)
- Dashboard par branch wise wasooli aur baqaya

4.9 Purchase aur Suppliers

- Supplier ka record, maal ki kharid (kitna, kis qeemat par)
- Purchase se stock khud barhta hai, supplier ka baqaya

4.10 Stock

- Stock sirf entries (`StockTransactions`) se: kharid, farokht, wapsi, adjustment
- Maujooda stock, kam stock ki ittila (ReorderLevel), gintee mein farq ki entry (wajah ke saath)

4.11 Returns

- Cash ya qist sale ki wapsi, product ke hisaab se
- Qist ho to baqaya kam, stock wapas, refund ka record

4.12 Roles aur Permissions

- Owner role banata hai aur har role ko ijazatein deta hai (jaise payment reverse, plan ki tabdeeli, stock adjust)
- User ko sirf wohi menu aur button dikhte hain jin ki ijazat ho

4.13 Shop Profile aur Multi-Shop

Dukaan ka naam kabhi code mein likha nahi hoga. Software ka koi apna fixed naam nahi, jo naam Shop Profile mein hoga wohi dikhega:

- Browser tab ka title, login page, menu ke upar
- Receipt, reports aur Rate List ka header (naam, logo, address, phone, NTN/registration)
- Receipt ka footer (jaise "Shukriya, qist waqt par ada karein")
- Owner ye sab kabhi bhi Settings se badal sakta hai, code ya dobara install ke baghair

Multiple shops ke liye do alag surtein hain:

Surat	Matlab	Kaise
Ek malik, kai shakhain	Ek hi business ki kai branches	`Branches` aur `Warehouses` (6.2), ek hi database, ek hi hisaab
Alag alag dukandar	Ye software kai alag dukandaron ko dena	**Har dukan ka apna database** (same code, alag `QistDB_<shop>`), is se ek dukan ka data doosri ko kabhi nazar nahi aata

Doosri surat ke liye do tareeqe hain:

1. **Alag install:** har dukan ka apna server aur apna database (sab se aasan aur mehfooz)
2. **Ek hosted server, har dukan ka alag database:** ek chhota `QistMaster` database jis mein `Shops` (code, name, connection string) hota hai. Login par dukan ka code likhte hain aur app us dukan ke database se judti hai. Naya dukandar add karna = naya database banao + ek row.

Sab dukanon ka data **ek hi database mein mila kar (TenantId se)** rakhna main **nahi** suggest karta: har table aur har SP mein TenantId ki jaanch lagani parti hai, aur ek galti se ek dukan ka hisaab doosri ko nazar a sakta hai.

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5. Business Rules (Hisaab ke usool)

5.1 Bunyadi usool

1. **Kul qeemat** = Plan ke **sab products ki qeemat ka jama** + Markup (markup ke tareeqe neeche 5.2 mein)
2. **Baqaya raqam** = Kul qeemat - Advance
3. **Qist ki raqam** = Baqaya raqam / Qiston ki tadaad, phir settings ke mutabiq gol (round) hogi. Rounding ka farq **aakhri qist** mein adjust hoga. Staff chahe to qist ki raqam khud likh sakta hai, tab qiston ki tadaad khud nikalegi (ya aakhri qist chhoti/barhi hogi).

4. **Due date** frequency ke hisaab se:
    - **Weekly:** har 7 din baad
    - **Fortnightly:** har 15 din baad
    - **Monthly:** har mahine usi tareekh ko (5 Jan, 5 Feb...). Agar mahine mein wo din na ho (jaise 31) to mahine ka aakhri din
    - **Custom:** har N din baad
    - Preview mein har qist ki date khud badli ja sakti hai (jaise Eid ke mahine mein aage karna)
5. **Payment allocation:** jo raqam wasool ho wo sab se purani unpaid qist mein pehle lagti hai, phir agli mein.
6. **Qist ka status:**
    - `Pending` : poori wasool nahi hui
    - `Paid` : poori wasool
    - `Late` : **calculate hota hai**, jab due date (aur grace days) guzar jayen aur qist `Pending` ho. Ye database mein store nahi hota.
7. **Plan ka status:** `Active` jab tak baqaya hai, `Completed` jab baqaya zero ho jaye.
8. Wasool shuda payment **delete nahi** hogi, ghalti ho to reverse entry banegi (hisaab saaf rakhne ke liye). Sirf Owner reverse kar sakta hai.
9. Late fee (agar settings mein on ho) har late qist par ek martaba lagegi.
10. **Jaldi settlement:** customer baqaya jaldi de de to Owner chhoot (discount) de sakta hai. Plan `Completed` ho jata hai.
11. **Plan cancel:** sirf Owner, wajah likhna zaroori. Maal wapas ho to stock barhta hai, wasool shuda raqam ka hisaab (wapsi ya katauti) alag note hoga.
12. **Reschedule:** baqaya qiston ka naya schedule banta hai, purani qistein history mein rehti hain.
13. **Payment baqaya se zyada** nahi ho sakti.
14. Har important kaam (reverse, cancel, discount, edit) `AuditLog` mein likha jayega.
15. **Wasooli ka rabta:** late customer se baat ka note aur "kab dega" ki tareekh `FollowUps` mein, aur dashboard par "aaj ka wada" dikhe.
16. **Multi-product plan:** sab products ka **ek hi plan, ek hi qist schedule aur ek hi baqaya** hota hai. Receipt aur plan par har product ki alag line dikhti hai. Plan ka ek product wapas ho to sirf us ki qeemat aur markup baqaya se kam hoti hai (`usp_SaleReturn_Create`), baqi products chalte rehte hain.
17. Stock har item ki tadaad ke hisaab se `StockTransactions` mein entry se kam/wapas hota hai.
18. **Plan mein baad ki tabdeeli** customer ki marzi se ho sakti hai (naya product, qist ka naya schedule, date, adjustment). Tafseel 5.5 mein.

5.2 Markup ke tareeqe

Type	Kaise lagta hai	Misal (Price 100,000, Advance 20,000, 10 qistein)
None	Koi markup nahi	Markup = 0
Fixed	Seedhi raqam	Markup = 12,000

Percent	Percentage ek martaba	15% par markup = 12,000
PercentPerMonth	Percentage har mahine (qiston ke mahine ginti se)	2% x 10 mahine = 20% par markup = 16,000
ManualTotal	Staff khud final kul qeemat likhta hai, markup khud nikalta hai	Kul 115,000 likha to markup = 15,000

- Percentage kis raqam par lage ye Settings mein hai: **poori qeemat** par ya **baqaya (qeemat - advance)** par. Default: baqaya.
- Preview mein markup ki akhri raqam bhi **badli ja sakti hai**. Aisa karne par type `ManualTotal` ho jata hai taake record mein sach likha rahe.
- Markup ki raqam aur kul qeemat receipt/plan par saaf likhi hogi.
- **Multi-product plan mein:** ye tareeqe poore plan ki jama qeemat par lagte hain. Agar rules (5.4) se auto markup lag rahi ho to **har product ki apni rule** lagti hai (jaise Oppo 0% aur Haier 15%) aur sab ka jama plan ki markup banta hai. Agar staff poore plan ki markup manual badal de to wo raqam products par **qeemat ke tanasub se** baant di jati hai, taake har item ka hisaab (aur return) theek rahe.

5.3 Live calculation (screen par)

Naya Plan page par chaar cheezein aapas mein judi hain: **markup, kul qeemat, qist ki raqam, qiston ki tadaad**. Jo bhi badlein, baqi khud update hoti hain. Calculation ek hi jagah (SP `usp_Sale_Preview`) mein hogi, JS sirf dikhayegi, taake screen aur database ka hisaab hamesha ek jaisa rahe.

5.4 Markup Rules (Naeem Electronics ke andaaz mein)

Public rate lists dekhne par ye andaaz samne aata hai: markup **brand/category ke hisaab se alag** hoti hai, aur **plan ki muddat** (3, 6, 12 mahine) ke saath badal sakti hai. Customer ko rate list mein product ki kul qeemat aur mahana qist likhi milti hai. Software mein isi tarah rakha jayega:

1. **Rules table (MarkupRules):** har brand ya category ki, har muddat ke liye alag percentage. Misal ke taur par (sirf samjhane ke liye, asli numbers aap tay karein):

Brand / Category	3 mahine	6 mahine	12 mahine
Oppo	0%	0%	0%
Redmi	5%	10%	15%
Samsung	0%	5%	10%
Default (baqi sab)	5%	10%	15%

2. **Konsi rule lagegi (tarteeb):** pehle Product ki apni rule, phir Brand, phir Category, aakhir mein Default. Offer ki date (`ValidFrom/ValidTo`) guzar jaye to rule khud band.
3. **Naya Plan par:** product aur muddat chunte hi markup % khud aa jata hai (`usp_Markup_Resolve`). Staff chahe to us ko badal sakta hai (5.2/5.3 ke mutabiq), aur plan mein likha rehta hai ke rule lagi ya manual.
4. **Percentage poori qeemat par ek martaba** lagti hai (jaise 100,000 par 15% = 115,000 kul), mahana nahi. Mahana wala tareeqa 5.2 mein alag option hai.
5. **Qist round-off:** misal: 115,000 / 12 = 9,583. 100 tak upar gol karne par qist 9,600 hui, aur aakhri qist

9,400 (farq wahan adjust). Agar dukaan ko "01" par khatam hone wali qist pasand ho (jaise 9,601) to Settings mein `RoundOffset = 1` kar dein.

6. **Rate List:** ye saari rules se khud ban jati hai (`usp_RateList_Generate`), taake dukaan ke bahar/WhatsApp par lagane ke liye print ya PDF mil sake. Markup badalte hi rate list bhi badal jati hai.

5.5 Plan mein baad ki tabdeeli (Amendment)

Customer ki marzi se plan mein baad mein bhi tabdeeli ho sakti hai. Bunyadi qaida: **jo ho chuka wo kabhi nahi badalta** (wasool shuda qistein, purane products ki markup). Sirf baqaya aur aane wali qistein naye sire se banti hain.

Tabdeeli	Kaise	Asar
Naya product add	Plan Detail > "Product add karo"	Naye product ki qeemat + markup baqaya mein jud jati hai, aane wali qistein dobara banti hain
Product wapas	Item par "Wapas"	Us ki qeemat + markup baqaya se kam
Qiston ki tadaad/muddat/raqam badalna	"Qistein dobara banao"	Baqaya wohi, qistein naye sire
Ek qist ki date aage/peechay	Qist par "Date badlo"	Sirf date badalti hai, raqam wohi
Chhoot ya extra raqam	"Adjust" (wajah ke saath)	Baqaya kam ya zyada
Jaldi settlement / cancel	Pehle se maujood	Section 5.1 ke rules

Rules

1. **Wasool shuda (Paid) qistein kabhi nahi chhui jatin.** Jis qist ki adhi raqam wasool hui hai uska baqi hissa naye schedule mein shamil hota hai.
2. Purani unpaid qistein **delete nahi hotin**, unka status `Superseded` ho jata hai aur nayi qistein banti hain. Is tarah history mein purana schedule bhi dikhta rehta hai.
3. **Naye product ki markup** aaj ki rule aur naye schedule ki muddat ke hisaab se lagti hai (ya manual). **Purane products ki markup wohi rehti hai**, dobara nahi nikalti.
4. Save se pehle **preview** dikhta hai: "pehle baqaya itna tha, ab itna hoga, nayi qistein ye hongi". Confirm karne par hi tabdeeli hoti hai.
5. Har tabdeeli par **wajah likhna zaroori** hai, aur wo `ContractAmendments` aur `AuditLog` mein record hoti hai.
6. **Kaun kar sakta hai:** jis role ko `Contract.Amend` permission mili ho (default: Owner).
7. Plan Detail aur receipt par **tabdeeli ki history** (kab, kya badla) dikhti hai.
8. **Completed ya Cancelled plan** mein tabdeeli nahi hoti. Customer naya saman le to naya plan banega.
9. Plan ki tabdeeli ke baad bhi payment allocation ka usool wohi rehta hai: sab se purani unpaid qist pehle.

5.6 Exact Formulas (SQL Specification)

Ye section stored procedures ka **final hisaab** hai. Har SP isi ke mutabiq likha jayega aur test hoga. Sab raqam **poore rupay** (PKR) mein, `ROUND(x, 0)` (0.5 upar jata hai). Jahan jama karna ho wahan aakhri hissa baqi farq uthata hai taake kul hamesha bilkul barabar rahe.

A. Sale ka hisaab

```
ItemsTotal      = SUM(Qty x UnitPrice)
ItemsNet        = ItemsTotal - DiscountAmount        (sale ke waqt ki chhoot, markup se PEHLE)
Markup          = (neeche B ke mutabiq)
OtherCharges    = file/processing charge (optional, Settings se on, default 0)
NetAmount       = ItemsNet + Markup + OtherCharges
FinancedAmount  = NetAmount - Advance                (Advance < NetAmount zaroori)
TotalInstallmentReceivable (TIR) = FinancedAmount
```

B. Markup

```
Base = ItemsNet                      agar MarkupBase = FullPrice
Base = ItemsNet - Advance            agar MarkupBase = Balance
FrequencyDays = 7 (Weekly) | 15 (Fortnightly) | 30 (Monthly) | IntervalDays (Custom)
TenureMonths  = CEILING(InstallmentCount x FrequencyDays / 30.0)

Fixed            : Markup = MarkupValue
Percent          : Markup = ROUND(Base x MarkupValue / 100, 0)
PercentPerMonth  : Markup = ROUND(Base x MarkupValue / 100 x TenureMonths, 0)
ManualTotal      : Markup = ManualTotal - ItemsNet            (>= 0 zaroori)
Rule (auto)      : har item ki apni markup, sab ka jama plan ki Markup
                   Markup_i = ROUND(ItemBase_i x Pct_i / 100, 0)
                   ItemBase_i = ItemNet_i                      (FullPrice)
                   ItemBase_i = ItemNet_i x (ItemsNet - Advance) / ItemsNet   (Balance)
                   (rounding ka farq aakhri item mein)
Plan ki markup manual badli jaye to items par qeemat ke tanasub se baant di jati hai
```

C. Auto schedule (N qistein)

```
A    = CEILING(FinancedAmount / N / RoundTo) x RoundTo + RoundOffset     (RoundTo = 0 ho to ROUND(
Qist 1 .. N-1 = A
Qist N        = FinancedAmount - (N-1) x A
Agar Qist N <= 0:  N' = CEILING(FinancedAmount / A), aakhri = FinancedAmount - (N'-1) x A
                   (qiston ki tadaad kam hoti hai, preview mein saaf likha aata hai)
```

Misal: Financed 115,000, N = 12, RoundTo = 100 => A = 9,600, pehli 11 qistein 9,600, aakhri 9,400. `RoundOffset = 1` par A = 9,601 aur aakhri 9,389.

Due dates (hamesha StartDate se ginte hain, pichli date se nahi, is liye drift nahi hota):

```
Weekly      : StartDate + 7  x (k-1)
Fortnightly : StartDate + 15 x (k-1)
Monthly     : DATEADD(MONTH, k-1, StartDate)    (31 Jan, 28 Feb, 31 Mar...)
Custom      : StartDate + IntervalDays x (k-1)
```

D. Custom amounts aur custom dates (sakht usool)

Staff har qist ki **raqam aur date** khud likh sakta hai. Misal (Financed = 45,000):

Qist	Raqam
1	5,000
2	8,000
3	12,000
4	20,000
Jama	**45,000**

SP ye sab check karta hai, ek bhi fail ho to **poora save rad**:

1. `SUM(Custom Installment Amounts) = TotalInstallmentReceivable` **bilkul barabar** (ek rupay bhi kam ya zyada nahi). Warna ghalti: "1,000 kam hain" ya "500 zyada hain".
2. Har raqam `> 0` aur poore rupay.
3. Har `DueDate >= StartDate` aur tarteeb se barhti hui (do qiston ki date barabar na ho).
4. Kam az kam 1 qist.

Screen par live "Baqi / Zyada: X" dikhta hai, aur ek button "aakhri qist mein baqi raqam daal do" jo aakhri raqam ko `FinancedAmount - (baqi sab qiston ka jama)` kar deta hai. Tabdeeli (reschedule/add items) mein bhi yehi usool hai, sirf total ki jagah **baqaya** (neeche E).

E. Baqaya aur payment allocation

```
Remaining_i   = Amount_i - PaidAmount_i - WaivedAmount_i      (sirf Status = Pending)
Outstanding   = SUM(Remaining_i)
Late (qist)   = Remaining_i > 0  AND  Today > DueDate + GraceDays
```

Payment `P` ka allocation (ek transaction mein, contract par lock):

```
1. P > Outstanding + LateFeeDue ho to REJECT (zyada raqam nahi)
2. Agar LateFeeFirst = 1:  pehle late fee, sab se purani qist se (LateFeePaid barhta hai)
3. Phir qiston mein, DueDate ki tarteeb se (sab se purani pehle):
      pay_i = MIN(Remaining_i, P_baqi);  PaidAmount_i += pay_i;  P_baqi -= pay_i
   (aane wali qiston ki pehle se ada bhi isi se: P_baqi bachi ho to agli qist mein lagti hai)
4. Har pay_i ki PaymentAllocations mein row
5. Remaining_i = 0 => Status = Paid;  Outstanding = 0 aur LateFeeDue = 0 => Contract Completed
```

F. Late fee

```
LateFeeType = None        : 0 (shuru mein yehi)
FixedOnce                 : Fee = LateFeeValue                          (har late qist par ek mar
PercentOnce               : Fee = ROUND(Remaining_i x LateFeeValue / 100, 0)
PerDay                    : Fee = LateFeeValue x DaysLate               (DaysLate = Today - DueDat
Fee = MIN(Fee, MaxLateFee)  agar MaxLateFee set ho
LateFeeDue_i = LateFee_i - LateFeePaid_i
```

Late fee SP payment ke waqt (aur due-list dekhte waqt) calculate karta hai, sirf **barhta** hai, kabhi ghat'ta nahi. Ye **kabhi Financed amount ya qiston ke jama mein shamil nahi hoti.** Maaf karna sirf `Payment.WaiveLateFee` permission se, wajah ke saath, audit mein.

Payment.WaiveLateFee permission se, wajah ke saath, audit mein.

G. Reverse

Payment reverse par us ki **har allocation ulti** jati hai (`PaidAmount_i -= alloc_i`, `LateFeePaid` bhi), payment `IsReversed = 1`, contract `Completed` tha to wapas `Active`. Reverse sirf `Payment.Reverse` permission se, wajah ke saath.

H. Settlement (jaldi hisaab)

```
SettlementAmount = Outstanding - SettlementDiscount + LateFeeDue(agar maaf na ho)
0 <= SettlementDiscount <= Outstanding
```

Chhoot **aakhri qiston se peeche ki taraf** `WaivedAmount` mein jati hai, baqi qistein poori wasool hokar `Paid`, contract `Settled`.

I. Plan ki tabdeeli (5.5) ke formulas

```
Reschedule : jis qist ki adhi raqam wasool hui, us ka Amount = PaidAmount kar ke Paid kar dete hai
             baqi hissa Outstanding mein.  Naya schedule: SUM(naye) = Outstanding
AddItems   : NewItemsNet = naye items (qeemat + markup, B ke formula, tenure = naye schedule ki)
             NewOutstanding = Outstanding + NewItemsNet - ExtraAdvance
             SUM(naya schedule) = NewOutstanding
Adjust     : NewOutstanding = Outstanding + (+/-)AdjustAmount  (>= 0)
```

J. Wapsi (Return)

```
Credit        = ReturnQty x UnitPrice + MarkupAmount x ReturnQty / Qty        (wapas item par mark
PaidTotal     = Advance + SUM(PaidAmount)
NewNet        = NetAmount - Credit
Refund        = MAX(PaidTotal - NewNet, 0)
NewOutstanding= MAX(NewNet - PaidTotal, 0)      => schedule: SUM(naya) = NewOutstanding
```

Stock mein `SaleReturn` entry (`QtyIn`), `SaleItems.ReturnedQty += ReturnQty`.

K. Cancel aur Stock

- **Cancel:** `Refund = PaidTotal - CancellationCharge` (Owner likhta hai, `>= 0`), sab saman stock mein wapas, contract `Cancelled`.
- **Stock:** Sale par `QtyOut = Qty`. `Balance = SUM(QtyIn - QtyOut)`. Balance `< Qty` ho to sale reject, jab tak `AllowNegativeStock = 1` na ho.
- **Cash sale:** `Payment = NetAmount` ek hi martaba (`PaymentType = CashSale`).

5.7 Costing aur Profit/Loss

Costing ka tareeqa: Weighted Average (har product aur warehouse ka alag `AvgCost`, `StockBalances` mein). Owner pehle isay manzoor kare.

```
Purchase par:  UnitCostNet_i = (LineTotal_i - discount ka hissa + charges ka hissa) / Qty_i
               NewAvg = (QtyOnHand x OldAvg + Qty x UnitCostNet) / (QtyOnHand + Qty)
Sale par:      SaleItems.UnitCost = us waqt ka AvgCost (sale ke saath jam jata hai), AvgCost nahi
Sale return:   stock wapas usi SaleItems.UnitCost par (COGS bilkul ulta hota hai)
Purchase return: stock PurchaseItems ki cost par kam, AvgCost dobara nikalta hai
```

```
Transfer:      bhejne wale ke AvgCost par jata hai, lene wale ka AvgCost dobara nikalta hai
```

SaleItems mein `UnitCost` column hoga taake COGS hamesha asli lagi hui cost se nikle.

Profit/Loss (default: accrual, yani sale ki tareekh par)

```
NetSales      = Sales.NetAmount (Active) - Returns ka Credit        (cash aur qist alag dikhein)
COGS          = SUM(Qty x UnitCost) - wapas hui items ki cost
GrossProfit   = NetSales - COGS                                     (qist ki markup NetSales mein
NetProfit     = GrossProfit - Expenses - BadDebtWriteOff + LateFeeCollected (+/- dusri configured
```

Ye ahem alag-alag cheezein hain aur reports mein alag label hongi:

- **Collection (naqd wasooli)** = us period mein asal mein mili raqam, sales nahi.
- **Outstanding qist (receivable)** = baqaya, naqd nahi aur profit nahi.
- **Purchases** kharcha nahi, stock hain jab tak wo bik na jayen.
- Qist ki markup ka faida sale par hi gin liya jata hai, is liye jo qist wasool na ho sake uske liye `BadDebtWriteOff` (Owner ki permission se, wajah ke saath) profit se minus hota hai.
- Agar owner "sirf wasool hone par faida" (cash basis) chahe to ye ek alag report ho sakti hai, default nahi.

---

6. Database Design (MS SQL Server)

Database ka naam `QistDB`. Paison ke columns `decimal(18,2)`. Database **SQL scripts** se banega (EF migrations nahi, section 9 dekhein).

6.1 Har table ke standard columns

- **Master/transaction tables:** `CreatedAt`, `CreatedBy`, `ModifiedAt`, `ModifiedBy` (sab tables mein, neeche har table mein dobara nahi likhe)
- **Master tables** (Customers, Products, Suppliers...) mein `IsDeleted` bhi (soft delete, record kabhi asal mein nahi mitta)
- **Sirf-jama tables** (Payments, StockTransactions, AuditLog, PaymentAllocations) mein `ModifiedAt/ModifiedBy` nahi: ghalti ho to **reverse entry** banti hai, purani row nahi badalti
- Concurrency ke liye ahem tables (Installments, InstallmentContracts, Payments) mein `RowVersion`

6.2 Security aur Setup

Table	Columns	Note
Branches	Id, Code, Name, Address, Phone, IsActive	Dukaan ki shakhain
Warehouses	Id, BranchId, Name, IsDefault	Har branch ke godam/showroom
Roles	Id, Name	Default: Owner, Staff, Accountant
Permissions	Id, Code, Module	Jaise `Sale.Create`, `Payment.Receive`, `Stock.Adjust`, `Report.Profit`

Permissions	Id, Code, Module	Jaise `Sale.Create`, `Payment.Reverse`, `Contract.Amend`, `Markup.Manage`, `Purchase.Create`, `Stock.Adjust`, `Report.View`, `Settings.Edit`
RolePermissions	RoleId, PermissionId	Kis role ko kya ijazat
Users	Id, Username, PasswordHash, FullName, BranchId, IsActive	
UserRoles	UserId, RoleId	Ek user ke ek ya zyada roles
PaymentMethods	Id, Name, Type (Cash/Bank/Wallet), IsActive	Cash, Bank, JazzCash, EasyPaisa...
NumberSequences	Id, BranchId, SeqType, Prefix, Year, LastNumber, Padding	SeqType: Receipt / Sale / Contract / Purchase / Return. Jaise `LHR-R-2026-000123`. `usp_Number_Next` se, lock ke saath
ShopProfile	Id, ShopName, LogoPath, Address, Phone, Email, RegistrationNo, ReceiptHeader, ReceiptFooter, CurrencySymbol, DefaultLanguage	Ek hi row. Owner kabhi bhi badal sakta hai, receipts/reports isi se naam lete hain
Settings	Key, Value	MarkupBase (FullPrice/Balance), RoundTo, RoundOffset, LateFeeType (None/FixedOnce/PercentOnce/PerDay), LateFeeValue, MaxLateFee, GraceDays, LateFeeFirst (0/1), AllowNegativeStock (0/1)
AuditLog	Id (bigint), UserId, Action, EntityName, EntityId, Details, CreatedAt	Kisne kya badla

6.3 Customers

Table	Columns	Note
Customers	Id, FullName, Phone, Phone2, Cnic, Address, Status (Active/Blacklisted)	Phone aur Cnic par index, Cnic unique
Guarantors	Id, CustomerId, FullName, Phone, Cnic, Relation, Address	
CustomerDocuments	Id, CustomerId, DocType, FilePath	File disk par, path DB mein

6.4 Products, Stock aur Purchase

Table	Columns	Note
Categories	Id, Name, TracksSerial, IsActive	Owner koi bhi category khud bana sakta hai (code mein fixed nahi)
Brands	Id, Name, IsActive	
Units	Id, Name	Piece, Set, Box...
Products	Id, Barcode, Name, CategoryId, BrandId, UnitId, CashPrice, MinSalePrice (null), WarrantyMonths, TracksSerial, ReorderLevel, IsActive	Stock yahan nahi, `StockBalances` aur `StockTransactions` se. Serial/IMEI ki uniqueness sirf tab jab TracksSerial on ho
Suppliers	Id, Name, Phone, Address, OpeningBalance	
Purchases	Id, PurchaseNo, SupplierId, BranchId, WarehouseId, PurchaseDate, TotalAmount, PaidAmount, Status	Maal ki kharid
PurchaseItems	Id, PurchaseId, ProductId, Qty, UnitCost, LineTotal	
StockBalances	ProductId, WarehouseId, QtyOnHand, AvgCost, RowVersion	Ledger ka khulasa, sirf SP update karta hai (lock ke saath). Ek report ledger aur balance ka farq pakarti hai
StockTransactions	Id, ProductId, WarehouseId, TxnType, QtyIn, QtyOut, UnitCost, RefType, RefId, TxnDate, Note	TxnType: Purchase / Sale / SaleReturn / PurchaseReturn / Adjustment / Transfer. **Stock = jama(QtyIn - QtyOut)**, kabhi seedha edit nahi hota

6.5 Sales aur Installment Contracts

Sales aur InstallmentContracts alag kyun: `Sales` bill/invoice hai (kya becha, kitne mein, kis ko, cash ya qist). `InstallmentContracts` sirf qist ka ma'ahida hai (advance, kitni qistein, kis waqfe se). Is se cash sale bhi isi system mein aati hai, aur qist ke ma'ahide ki tabdeeli bill ko nahi chhoo-ti.

Table	Columns	Note
Sales	Id, SaleNo, BranchId, WarehouseId, CustomerId (cash mein null ho sakta hai), SaleType (Cash/Installment), SaleDate, ItemsTotal, MarkupAmount, DiscountAmount, OtherCharges, NetAmount, Status (Active/Cancelled)	NetAmount = ItemsTotal - DiscountAmount + MarkupAmount + OtherCharges
SaleItems	Id, SaleId, ProductId, Qty, UnitPrice, LineTotal, MarkupPercent, MarkupAmount, MarkupRuleId (null), SerialNo, WarrantyMonths, ReturnedQty	UnitPrice us waqt ki, baad mein product ki qeemat badle to ye nahi badalti

InstallmentContracts	Id, ContractNo, SaleId (unique), CustomerId, GuarantorId (null), BranchId, MarkupType, MarkupValue, Advance, FinancedAmount, InstallmentCount, FrequencyType, IntervalDays (null), InstallmentAmount, StartDate, SettlementDiscount, Status	MarkupType: Rule / None / Fixed / Percent / PercentPerMonth / ManualTotal. Status: Active / Completed / Settled / Cancelled
Installments	Id, ContractId, InstallmentNo, DueDate, Amount, PaidAmount, WaivedAmount, LateFee, LateFeePaid, AmendmentId (null), Status	Status: Pending / Paid / Superseded. Baqi = Amount - PaidAmount - WaivedAmount. **Late stored nahi hota**, calculate hota hai
Payments	Id, ReceiptNo, BranchId, ContractId (null), SaleId, PaymentMethodId, PaymentType (Advance/Installment/Settlement/CashSale/LateFee), Amount, PaymentDate, Note, IsReversed, ReversedBy, ReversedAt, ReverseReason	
PaymentAllocations	Id, PaymentId, InstallmentId, Amount	Payment kin qiston mein lagi
ContractAmendments	Id, ContractId, AmendmentType, OldBalance, NewBalance, Reason, Details (JSON)	AddItems / ReturnItem / Reschedule / ChangeDate / Adjust
FollowUps	Id, ContractId, ContactDate, Note, PromiseDate (null)	Wasooli ka rabta
MarkupRules	Id, ScopeType (Product/Brand/Category/Default), ScopeId (Product/Brand/Category ka Id), TenureMonths, MarkupPercent, IsActive, ValidFrom, ValidTo	Section 5.4

6.6 Returns

Table	Columns	Note
SaleReturns	Id, ReturnNo, SaleId, ContractId (null), ReturnDate, Reason, TotalAmount, RefundAmount, Status	Cash aur qist dono sale ki wapsi
SaleReturnItems	Id, ReturnId, SaleItemId, Qty, Amount	Wapsi par SaleItems.ReturnedQty barhta hai, StockTransactions mein SaleReturn entry,

contract ho to baqaya kam

6.6A Supplier payments, Expenses aur Cash/Bank

Table	Columns	Note
SupplierPayments	Id, PaymentNo, SupplierId, PurchaseId (null), PaymentMethodId, Amount, PaymentDate, Note, IsReversed	Supplier ko raqam di
PurchaseReturns	Id, ReturnNo, PurchaseId, ReturnDate, Reason, TotalAmount	Supplier ko maal wapas
PurchaseReturnItems	Id, ReturnId, PurchaseItemId, Qty, Amount	Stock `PurchaseReturn` entry se kam
ExpenseCategories	Id, Name, IsActive	Kiraya, bijli, tankhwah... (owner khud banata hai)
Expenses	Id, BranchId, CategoryId, ExpenseDate, Amount, PaymentMethodId, Note, AttachmentPath (null), Status	Net profit mein minus hota hai
CashBankAccounts	Id, BranchId, Name, Type (Cash/Bank/Wallet), OpeningBalance, IsActive	Har PaymentMethod ek account se juda
CashBankTransactions	Id, AccountId, TxnDate, AmountIn, AmountOut, RefType, RefId	Sirf-jama ledger: payments, supplier payments, expenses, refunds se khud banti hai
RefreshTokens	Id, UserId, TokenHash, ExpiresAt, RevokedAt (null)	Section 10
IdempotencyKeys	RequestKey, Endpoint, UserId, ResultRef, CreatedAt	Unique (RequestKey, Endpoint)

6.7 Relationships

```
Branches 1---* Warehouses
Roles *---* Permissions (RolePermissions)      Users *---* Roles (UserRoles)
Suppliers 1---* Purchases 1---* PurchaseItems *---1 Products
Customers 1---* Sales 1---* SaleItems *---1 Products
Sales 1---0..1 InstallmentContracts 1---* Installments
InstallmentContracts 1---* Payments 1---* PaymentAllocations *---1 Installments
InstallmentContracts 1---* ContractAmendments, FollowUps
Sales 1---* SaleReturns 1---* SaleReturnItems *---1 SaleItems
Products 1---* StockTransactions *---1 Warehouses
```

6A. Stored Procedures

Faisla: poora data access SP se

EF Core istemal **nahi** hoga. Har kaam (simple CRUD bhi) stored procedure se hoga, C# mein Dapper se call. **Hisaab ka har usool (markup, qist banana, payment lagana, baqaya, stock) sirf SP mein hai**, C# mein kahin dobara nahi likha jata.

CRUD SPs ka pattern

Har entity ke liye: `usp_<Entity>_Save` (naya ya edit), `usp_<Entity>_Get`, `usp_<Entity>_List` (search, paging ke saath), `usp_<Entity>_Delete` (soft delete). Jaise `usp_Customer_Save`, `usp_Product_List`.

Business SPs

SP	Kaam
`usp_Auth_Login`	User, password hash aur uski permissions wapas karta hai
`usp_Number_Next`	Receipt/Sale/Contract number (branch aur saal ke hisaab se, lock ke saath)
`usp_Markup_Resolve`	Har product ki brand/category/muddat dekh kar markup % (5.4 ka tarteeb)
`usp_Sale_Preview`	Items, markup, advance, frequency se schedule nikal kar dikhana (save nahi). Qist ki raqam ya count dono mein se ek input
`usp_Sale_Create`	**Transaction:** Sale + SaleItems + StockTransactions (Sale). Installment sale ho to InstallmentContract + final schedule (TVP) + advance Payment bhi, ek hi transaction mein. Check: qiston ka jama = baqaya
`usp_Payment_Receive`	**Transaction:** receipt no., sab se purani unpaid qist se allocate, PaidAmount, plan complete check, partial payment
`usp_Payment_Reverse`	Allocations wapas, payment reverse mark, audit
`usp_Contract_Settle`	Jaldi poora hisaab, chhoot ke saath
`usp_Contract_Cancel`	Plan cancel, stock wapas, audit
`usp_Contract_Reschedule`	Baqaya wohi, qistein naye sire (5.5)
`usp_Contract_AddItems`	Purane plan mein naye products: SaleItems, stock, baqaya barhana, unpaid qistein `Superseded`, nayi qistein, `ContractAmendments`
`usp_Contract_Adjust`	Chhoot ya extra raqam, qistein naye sire
`usp_Installment_ChangeDate`	Ek qist ki date badalna
`usp_ContractAmend_Preview`	"Pehle ye tha, ab ye hoga" dikhana, save nahi
`usp_SaleReturn_Create`	Wapsi: ReturnedQty, StockTransactions, contract ho to baqaya kam aur schedule nave sire, refund

`usp_Purchase_Create`	Purchase + PurchaseItems + StockTransactions (Purchase), supplier ka hisaab
`usp_Stock_Adjust`	Gintee mein farq ya kharabi ki entry (wajah zaroori)
`usp_Stock_Balance`	Product/warehouse wise maujooda stock
`usp_Installment_DueToday`, `usp_Installment_Late`	Aaj ki due aur late qistein
`usp_Dashboard_Summary`	Dashboard ke sab numbers ek call mein
`usp_RateList_Generate`	Rules se Rate List
`usp_Report_DailyCollection`, `_Monthly`, `_Defaulters`, `_CustomerLedger`, `_StockValuation`, `_PurchaseSummary`, `_Profit`	Reports

SP likhne ke usool

- **Naam `usp_` se shuru**, kabhi `sp_` se nahi: SQL Server `sp_` wale naam pehle `master` database mein dhoondta hai (sust aur ghalat SP chalne ka khatra).
- Sale, Payment, Return, Settle, Cancel, Purchase ke SPs `@RequestKey uniqueidentifier` lete hain (dobara submit se bachao, section 10).
- Har paison wale SP mein: `SET XACT_ABORT ON`, `BEGIN TRAN`, `TRY/CATCH`, ghalti par `ROLLBACK` aur `THROW`.
- Payment aur contract ki tabdeeli mein contract par `UPDLOCK, HOLDLOCK`, taake do staff ek qist do baar wasool na kar saken.
- Input ki jaanch SP mein bhi (raqam > 0, plan active, payment baqaya se zyada na ho, stock kaafi ho).
- Late status hamesha `DueDate < today AND Status = Pending` se calculate, column mein store nahi.
- Tareekhen PKT (UTC+5) mein; SP mein ek hi function se "aaj ki tareekh" lein.
- Har SP par app ke SQL login ko sirf `EXECUTE` ki ijazat (table par seedhi nahi, section 10).

Scripts ka intezam

- `Database/Tables/`, `Database/StoredProcedures/`, `Database/Seed/` mein numbered `.sql` scripts (`CREATE OR ALTER PROCEDURE`).
- Ek chhota script runner (jaise **DbUp** package ya khud ka) jo scripts ko tarteeb se chalata hai aur `SchemaVersions` table mein likhta hai ke kaun si chal chuki.
- Sab scripts Git mein.

C# se call

```
// Dapper
var result = await conn.QueryFirstAsync<PaymentResult>(
    "usp_Payment_Receive",
    new { ContractId = id, Amount = amount, PaymentMethodId = m, UserId = userId },
    commandType: CommandType.StoredProcedure);
```

7. API Endpoints

Base URL: `/api`. Har endpoint ko permission ka code laga hota hai (jaise `Payment.Reverse`).

Group	Endpoints
Auth	`POST /auth/login`
Customers	`GET /customers?search=`, `GET /customers/{id}`, `POST`, `PUT`, `POST /customers/{id}/guarantors`, `GET /customers/{id}/ledger`
Products	`GET /products`, `POST`, `PUT`, `GET /products/{id}/stock`
Markup	`GET /markup/resolve?productId=&months=`, `GET/POST/PUT /markup-rules`, `GET /rate-list?brand=&category=`
Sales	`POST /sales/preview`, `POST /sales`, `GET /sales/{id}`, `GET /sales?search=`
Contracts	`GET /contracts/{id}`, `POST /contracts/{id}/settle`, `/cancel`, `/reschedule`, `/items`, `/adjust`, `/amend-preview`, `GET /contracts/{id}/amendments`, `GET/POST /contracts/{id}/followups`
Installments	`GET /installments/due-today`, `GET /installments/late`, `POST /installments/{id}/change-date`
Payments	`POST /payments`, `GET /payments/{id}/receipt`, `POST /payments/{id}/reverse`
Returns	`POST /sales/{id}/returns`, `GET /returns/{id}`
Purchases	`GET/POST /purchases`, `GET/POST /suppliers`
Stock	`GET /stock`, `POST /stock/adjust`, `GET /stock/transactions`
Reports	`GET /reports/daily`, `/monthly`, `/defaulters`, `/stock-valuation`, `/purchases`, `/profit`, `GET /dashboard/summary`
Admin	`GET/PUT /shop-profile`, `GET/PUT /settings`, `GET/POST/PUT /users`, `/roles`, `/branches`, `/warehouses`, `/payment-methods`

Response: JSON. Error par proper HTTP code (400, 401, 403, 404) aur saaf message.

---

8. Frontend Pages

Technology: HTML5 + Bootstrap 5 + CSS, aur **jQuery + AJAX** (`$.ajax`). Ek `api.js` mein ek hi `$.ajax` wrapper hoga jo token lagata hai, loading/ghalti ka saaf message dikhata hai, aur 401 par login par bhejta hai.

Page	Kaam
`login.html`	Login
`index.html`	Dashboard

`customers.html, customer.html`	List, search, detail (plans, khata)
`products.html`	Products, brand/category
`new-sale.html`	Cash ya Installment sale, cart, markup, preview, schedule edit
`contract.html`	Plan detail, qistein, items, tabdeeli ke buttons, history
`payments.html`	Qist wasooli (search aur jama)
`receipt.html`	Print / WhatsApp
`returns.html`	Wapsi
`purchases.html, suppliers.html`	Kharid aur suppliers
`stock.html`	Stock, adjustment, transactions
`followups.html`	Late customers, "aaj ka wada"
`markup-rules.html, rate-list.html`	Markup rules aur Rate List
`reports.html`	Reports
`shop-profile.html, settings.html, users.html, roles.html, branches.html`	Admin

User-friendly usool

- Bade buttons, saaf font, kam typing
- Har page par upar ek **phone number search box**
- Late qistein laal, aaj ki due peeli, paid hari
- Mobile aur computer dono par theek (Bootstrap responsive)
- Har kaam ke baad saaf message
- Delete/reverse/cancel se pehle confirmation
- User ko sirf wohi menu/button dikhein jis ki permission ho

---

9. Architecture aur Project Structure

Architecture

```
Bootstrap + HTML + CSS
        +
jQuery + AJAX
        ↓
ASP.NET Core 10 Web API
        ↓
Controllers            (request lena, permission check, jawab dena)
        ↓
Services               (orchestration: SP ko call karna, mapping)
```

```
        ↓
Dapper Repository      (SP chalane ka wrapper)
        ↓
SQL Stored Procedures  (poora hisaab aur business logic)
        ↓
SQL Server
```

Zimmedari ka taqseem

Layer	Karta hai	Nahi karta
Controller	Input lena, permission check, HTTP code	Hisaab
Service	SP call karna, ek se zyada SP ki tarteeb, DTO mapping, request ki shakal ki jaanch	**Qist banana, markup, payment allocation, baqaya, stock: kuch nahi**
Repository	Dapper se SP chalana, parameters lagana	Koi logic
SP	**Sab financial aur business logic, transaction, validation**	

Usool: Calculation / financial logic sirf SP mein. C# SP ko call kare aur nateeja wapas kare.

Project Structure

```
QistApp/
├── Controllers/
├── Services/            (orchestration, mapping)
├── Repositories/        (Dapper SP executor)
├── Models/ DTOs/
├── Security/            (JWT, permission policies)
├── Database/
│   ├── Tables/          (001_..., 002_... .sql)
│   ├── StoredProcedures/
│   ├── Seed/            (roles, permissions, payment methods)
│   └── SchemaVersions
├── wwwroot/
│   ├── css/  js/ (jquery, api.js, pages)  *.html
├── appsettings.json / appsettings.Production.json
└── Program.cs
```

---

10. Security

- Password hashing (`PasswordHasher`), JWT access token **chhoti muddat (30-60 min)** ka, aur uske saath **refresh token** (DB mein hash ke taur par, `RefreshTokens` table, 8-12 ghante). User band karne ya role badalne par uske refresh tokens radd ho jate hain, is liye asar foran (access token ki muddat tak) hota hai. *Marhala 1 ke code mein abhi 8 ghante ka ek token hai, ye marhala 2 mein refresh token se badla jayega.*
- **Dobara submit se bachao (Idempotency):** screen har form kholte hi ek naya `RequestKey` (GUID) banati hai aur submit par button band kar deti hai. Sale, Payment, Return, Settle, Cancel, Purchase ke SPs `@RequestKey` lete hain aur `IdempotencyKeys` table (unique Key + Endpoint) mein dekhte hain: key pehle aa chuki ho to naya record nahi banta, purana nateeja wapas milta hai. Is se double-click ya

net ki dobara koshish se ek hi payment do baar nahi lagti.

- **Permission based access:** har API par permission ka code, sirf Owner ya jis role ko ijazat ho
- **SQL login ki hadd:** app ka SQL login sirf stored procedures par `EXECUTE` kar sake, tables par seedha `SELECT/INSERT/UPDATE/DELETE` nahi. Is se agar app mein koi ghalti ho to bhi database mehfooz rahe
- Dapper mein hamesha parameters (SQL injection se bachao)
- Input validation (CNIC 13 digits, phone format, raqam > 0) frontend aur SP dono mein
- Login par koshishon ki had (rate limit)
- Connection string aur secrets code mein nahi, environment variable / User Secrets mein
- HTTPS, CORS sirf apne domain ke liye
- CNIC aur customer ki images ka data private, in tak sirf permission walon ki pohanch
- **Roz ka database backup** (SQL Server Agent job), aur hafte mein ek copy doosri jagah; **restore ko waqtan fawaqtan test karna**

---

11. Development Plan

Marhala	Kaam
1	Project setup, SQL scripts runner, tables, roles/permissions, login, Shop Profile, `usp_Number_Next`
2	Branches, Customers, Guarantors, Products, Payment Methods (CRUD SPs + pages)
3	Suppliers, Purchases, StockTransactions, stock page
4	Markup Rules, `usp_Markup_Resolve`, `usp_Sale_Preview`, `usp_Sale_Create` (cash + installment), Naya Sale page
5	`usp_Payment_Receive`, reverse, receipt
6	Dashboard, due/late, FollowUps
7	Returns, settle, cancel, reschedule, plan ki tabdeeli
8	Expenses, Cash/Bank, Reports (profit/loss), Rate List, export
9	Audit, SP testing (5.6 ki misalon ke test cases, galat/zyada raqam, ek hi waqt do payments), backup, production setup
10	(Baad mein) Urdu RTL, SMS/WhatsApp reminders

---

12. Future Improvements

- Urdu RTL interface
- Qist se pehle SMS / WhatsApp reminder
- Har piece ka alag serial/IMEI (`ProductSerials`)
- Multiple branches par ek markazi server aur branch ke darmiyan stock transfer
- Barcode scanner

- Android app

---

13. Assumptions (Mafroozat)

- Interface pehle Roman Urdu/English, Urdu RTL baad mein.
- Qistein hafta war, 15 din, mahana ya custom ho sakti hain (har plan par alag). Markup har plan par alag, ya bilkul na lage.
- Markup ke rules brand/category aur muddat ke hisaab se, staff manual badal sakta hai.
- Ek plan mein ek ya kai products, sab ki ek hi muddat aur ek hi schedule.
- Plan ki tabdeeli sirf `Active` plan mein, wajah ke saath, `Contract.Amend` permission walay user ke zariye.
- Currency PKR, time zone PKT.
- Dukan ka naam, logo aur receipt ki tafseel Shop Profile se customize hoti hai, code mein fixed nahi.
- Alag alag dukandaron ke liye har dukan ka alag database (data milaya nahi jata).
- Late fee shuru mein band (Settings se on).
- Customer ki images disk par, DB mein sirf path.
- Serial/IMEI shuru mein har item line par ek.
- Database SQL Server Developer Edition par banega aur test hoga. **Licensing:** Developer edition sirf development/testing ke liye muft hai, asli (production) istemal ke liye Express (muft) ya Standard (paid) license chahiye. Code dono par ek jaisa chalega.
- Markup ke rules ka andaaz Naeem Electronics ke public rate lists se liya gaya hai, asli numbers dukaan khud tay karegi.

---

14. Production Architecture

Deployment (ek dukaan, LAN)

```
Staff ke computers / mobiles (browser)
        ↓  (dukaan ka LAN / WiFi, HTTPS)
Server PC:  ASP.NET Core 10 app (Kestrel, Windows Service)  +  SQL Server
        ↓
Roz ka backup: dusri drive + cloud/USB copy
```

- **App:** Kestrel par, Windows Service ke taur par (computer chalte hi khud shuru). Chahein to IIS ke peeche.
- **SQL Server:** usi PC par ya alag; production mein Express/Standard license.
- **HTTPS:** LAN mein local certificate; agar internet se kholna ho to asli domain aur certificate.
- **Multi-branch:** markazi server (cloud VPS ya head office) aur branches internet/VPN se.

Multiple shops

- Har dukan ka **alag database**, same code (4.13). Scripts runner har dukan ke database par chalta hai.
- Hosted surat mein `QistMaster.Shops` se dukan ka connection string chuna jata hai, aur backup har dukan ka alag.

- Dukan ka naam, logo aur receipt ki sab cheezein us dukan ke `ShopProfile` se aati hain.

Configuration

- `appsettings.Production.json`, secrets environment variables mein
- Development aur Production ke database alag
- Time zone PKT

Reliability

- **Backup:** roz full backup, ahem waqt mein transaction log backup (Full recovery model ho to), hafte mein ek offsite copy, **mahine mein ek restore test**
- UPS aur surge protector server ke liye (bijli jane se database kharab na ho)
- Health check endpoint (`/health`) jo app aur database dono ko dekhe
- Logging (Serilog se file mein), ghalti par saaf message user ko, tafseel log mein
- Global error handler, kabhi stack trace user ko nahi

Update karne ka tareeqa

1. Backup lein
2. Naye SQL scripts chalayen (runner khud sirf nayi scripts chalata hai)
3. Naya app deploy karein
4. Chhoti jaanch (login, ek payment, ek report)
5. Masla ho to backup se wapas

Performance

- Index: Customers (Phone, Cnic), Installments (DueDate, ContractId, Status), Payments (PaymentDate), StockTransactions (ProductId, WarehouseId)
- List pages mein paging
- Dashboard ek SP call se

Data ki hifazat

- Soft delete, reverse entries, AuditLog: koi hisaab chupke se nahi badalta
- Purana data kam az kam 5-7 saal rakhna (qanooni/hisaab ke liye)

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15. Acceptance Checklist

Development ke baad ye sab pass hona zaroori hai:

1. Owner bina code badle nayi category, brand aur product bana sakta hai.
2. Cash sale stock kam karti hai aur unique invoice banati hai.
3. Qist sale ka preview sahi hai, aur advance aur baqaya ka hisaab 5.6 ki misalon se bilkul milta hai (115,000 / 12 qistein = 9,600 x 11 + 9,400).
4. Custom qistein: jama farq ho to save nahi hota; jama barabar ho (5,000 + 8,000 + 12,000 + 20,000 = 45,000) to hota hai.
5. Partial payment sahi qist mein lagti hai, customer ka baqaya sahi update hota hai, aur **ek hi submit do**

baar bhejne par doosri payment nahi banti.

6. Late status due date aur grace days ke mutabiq aata hai.
7. Purchase stock barhati hai aur supplier ka baqaya update karti hai; supplier payment se ledger update hota hai.
8. Return aur cancel purani history nahi mitate, stock aur baqaya sahi karte hain.
9. Weighted average cost sahi nikalti hai aur COGS sale ke waqt ki cost se banta hai.
10. Profit/loss mein sales, cost, expenses, asal wasooli aur outstanding alag alag aate hain.
11. Bina ijazat wala staff reverse, settle, cancel, discount ya plan ki tabdeeli nahi kar sakta.
12. User band karne par uska token ek access-token ki muddat ke andar kaam karna band kar deta hai.
13. Stock ledger aur `StockBalances` ka farq report mein zero aata hai.
14. Backup alag computer par restore hokar chalta hai (test).

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16. Out of Scope (Abhi shaamil nahi)

- Payroll/HR, manufacturing, advanced CRM, e-commerce, loyalty programme, AI features
- Bara enterprise accounting (sirf wo ledger/finance jo shop ke reports ke liye zaroori hai)
- Native mobile app (web UI mobile par theek chalegi)
- Tax/withholding: shuru mein configurable aur optional, dukan ke asal qanooni intezam ke mutabiq baad mein